

No. of Printed Pages : 4
Roll No.

220842

**4th Sem / Computer, Computer (For Speech and
Hearing Impaired)
Subject : Computer Organisation and Architecture**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What does RISC stand for?
- a) Reduced Instruction Set Computer
 - b) Random Instruction Set Computer
 - c) Real-time Instruction Set Computer
 - d) Read Instruction Set Computer
- Q.2 Which instruction format uses the least number of addresses?
- a) Three Address
 - b) Two Address
 - c) One address
 - d) Zero address
- Q.3 What does DMA stand for in memory management?
- a) Direct memory Access
 - b) Data Memory Access

- c) Dynamic Memory Allocation
- d) Direct Memory Allocation

Q.4 What does BIOS stand for?

- a) Basic Input/Output System
- b) Binary Input/Output System
- c) Basic Internal Operating System
- d) Binary Internal Operating System

Q.5 Which of the following is a form of parallel processing?

- a) Time-sharing
- b) Batch processing
- c) Pipelining
- d) None of the above

Q.6 Which of the following describes asynchronous data transfer?

- a) Data sent with a fixed clock
- b) Data sent without synchronization
- c) Data sent only in batches
- d) Data sent in real-time

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 In stack organization, data is accessed in a last-in, first-out (LIFO) manner. (True/False)

- Q.8 What does “Immediate addressing mode” mean?
- Q.9 What one difference between RAM and ROM?
- Q.10 What is the role of DMA in data transfer operations?
- Q.11 A crossbar switch is a type of interconnection network that allows multiple processors to communicate simultaneously. (True/False)
- Q.12 Handshaking is a method used to synchronize data transfer between devices. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is immediate addressing mode?
- Q.14 What is stack organization in a CPU?
- Q.15 Explain associative memory.
- Q.16 Write short note on virtual memory.
- Q.17 What is the function of BIOS?
- Q.18 What is synchronous data transfer?
- Q.19 Describe a crossbar switch.
- Q.20 Explain DMA data transfer technique.
- Q.21 Define I/O interface.
- Q.22 What is strobe control?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Define the various instruction formats : three-address, two address, one-address, and zero-address instructions.

Q.24 Write a short notes on:

a) magnetic disks

b) cache memory

Q.25 Explain the concept of pipelining in parallel processing and its impact on performance.